KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

MINI PROJECT REPORT



TRAFFIC SIGNAL CONTROL SYSTEM USING ANN AND FUZZY LOGIC

ETEG 425

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ABSTRACT

Growing traffic volumes and limited road capacity have made congestion a major urban problem. Fixed-timing traffic signals cannot adjust to real-time traffic, causing longer queues, wasted fuel, and higher emissions.

This project builds an intelligent traffic signal system combining an Artificial Neural Network (ANN) with a Per-Junction Fuzzy Logic Controller (FLC). The ANN predicts vehicle waiting time from traffic density, junction ID, and time of day. This prediction, along with current traffic density, is fed into a fuzzy controller that decides the green signal duration. Instead of using one fixed set of rules for every junction, the system generates separate membership functions for each junction based on its own traffic pattern, allowing better adaptation across different intersections.

The project also analyzes traffic patterns, evaluates the ANN using regression and classification metrics, visualizes fuzzy controller behavior through response curves and heatmaps, and compares the adaptive controller with a fixed-time controller through simulation. On a dataset of 48,120 records from four junctions, the ANN achieves an R^2 of 0.963 and 97.6% classification accuracy, while the adaptive controller reduces average waiting time by about 23% compared to a fixed 45-second signal.

Keywords: *Traffic Signal Control, Artificial Neural Network, Fuzzy Logic Controller, Intelligent Transportation System, Adaptive Traffic Signals, Machine Learning, Traffic Density, Smart City*

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INTRODUCTION

1.1 Background

Efficient traffic management is an important engineering problem in urban areas because vehicle demand continues to increase while road capacity remains limited. Conventional signal controllers usually assign fixed green durations without considering the present traffic load. Although this method is simple, it can produce unnecessary delay, longer queues, higher fuel consumption, and congestion during peak periods.

Recent developments in Artificial Intelligence have made adaptive traffic control more practical. Artificial Neural Networks (ANNs) can learn nonlinear traffic patterns from historical data, whereas Fuzzy Logic Controllers (FLCs) can convert uncertain traffic conditions into rule-based control decisions without requiring an exact mathematical traffic model.

In this project, both techniques are combined to develop an intelligent traffic signal control system. The ANN estimates vehicle waiting time using vehicle count, junction number, and hour of the day. The predicted waiting time and current traffic density are then supplied to the fuzzy controller to calculate a suitable green signal duration.

A key feature of the proposed method is the use of independent fuzzy controllers for individual junctions. Instead of applying one common traffic scale to all intersections, the membership functions are generated from each junction's own traffic distribution. This makes the controller more adaptive to local traffic behavior.

The complete workflow includes traffic pattern analysis, ANN model evaluation, fuzzy controller visualization, and comparison with a conventional fixed-time controller.

1.2 Objectives

The objectives of this project are:

- To develop an ANN model to predict vehicle waiting time at junctions.
- To design a fuzzy logic controller that uses this predicted waiting time and traffic density to set adaptive green signal timing.
- To evaluate its performance against a fixed-time traffic signal controller.

1.3 Problem Statement

Traditional traffic signals allocate fixed green durations without considering the current vehicle demand at an intersection. During heavy traffic, this can increase queue length and waiting time. During light traffic, the same fixed timing may waste green time even when fewer vehicles are present.

Several adaptive systems use fuzzy logic for signal timing, but many of them use one common set of membership functions for all junctions. Since each junction can have a different traffic range and peak-hour behavior, a single fuzzy scale may not represent all intersections accurately.

Therefore, an intelligent controller is required that can estimate traffic conditions and adjust green duration according to the characteristics of each individual junction.

METHODOLOGY

The proposed traffic signal control system uses a hybrid ANN-FLC methodology. The main stages are data preprocessing, exploratory traffic analysis, ANN-based waiting-time prediction, per-junction fuzzy control, adaptive signal timing, and performance evaluation.

2.1 Data Collection and Preprocessing

The dataset contains DateTime, Junction, and Vehicle Count attributes collected from four junctions, giving 48,120 hourly records. The data are first checked for missing values, duplicate entries, and invalid vehicle counts so that inconsistent records do not affect model training.

The Hour feature is extracted from DateTime to represent daily variation. Since measured waiting-time data are not available, waiting time is generated using traffic density and peak-hour behavior, with controlled random variation to make the simulated values closer to practical traffic conditions.

The processed dataset is then used for training the Artificial Neural Network and constructing the fuzzy controllers.

2.2 Traffic Pattern Analysis

Before model development, exploratory data analysis is performed to understand the main characteristics of the traffic dataset. The analysis includes:

- Distribution of traffic density.
- Hourly traffic variation, analyzed separately for each junction.
- Average ANN-predicted waiting time throughout the day, per junction.
- Junction-wise traffic statistics.

These analyses identify the traffic distribution, peak periods, and junction-wise differences. This information is important for designing an adaptive controller that responds to actual traffic variation rather than using a fixed timing pattern.

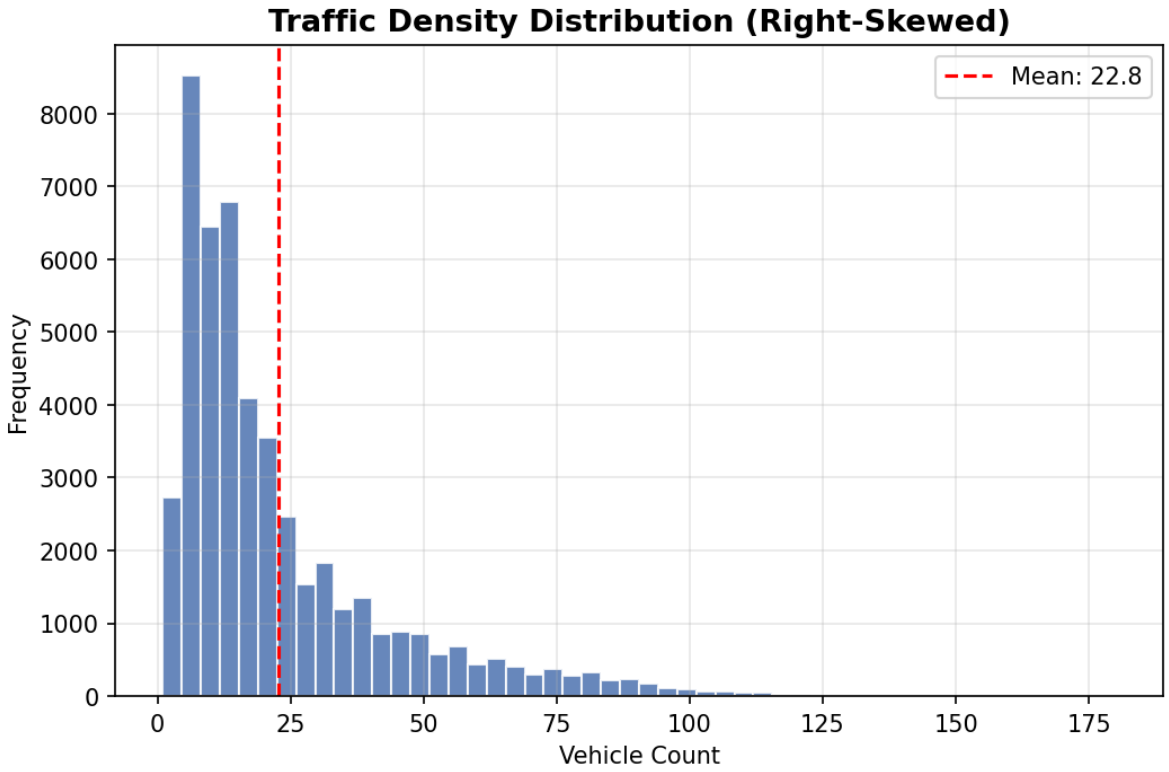


Figure 1: Traffic Density Function

The traffic density distribution is strongly right-skewed, with a skewness of 1.82 and a mean of about 23 vehicles per record. Most samples represent light or moderate traffic, while a smaller number of samples correspond to heavy congestion. This supports the use of quantile-based membership functions, because a single fixed scale would not represent both low-density and high-density conditions effectively.

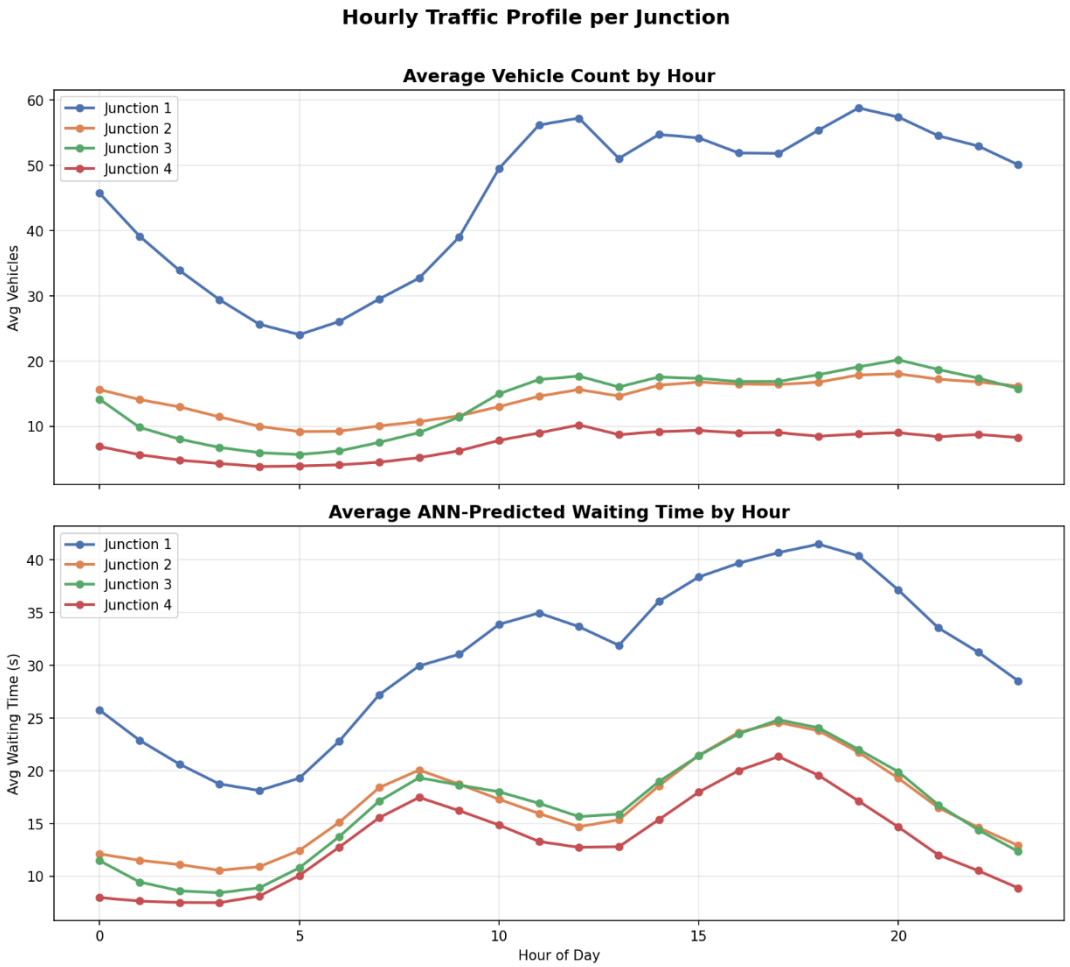


Figure 2: Hourly Traffic Profile per Junction

Figure 2 shows the hourly traffic profile separately for each junction. Junction 1 has a higher traffic volume than Junctions 2-4, while all junctions show morning and evening peak periods. The predicted waiting time follows the vehicle-count trend, indicating that the ANN has learned the density-waiting-time relationship in a consistent manner.

2.3 Artificial Neural Network for Waiting Time Prediction

An Artificial Neural Network (ANN) is used to estimate vehicle waiting time from the processed traffic data. The input features are:

- Vehicle Count
- Hour of the Day
- Junction Number

The output of the model is the predicted waiting time for each traffic record.

Before training, the input features are standardized so that no feature dominates due to its numerical range. The ANN uses three hidden layers with 64, 32, and 16 neurons, ReLU activation, and an 80/20 train-test split stratified by junction. Early stopping is applied to reduce overfitting and improve generalization.

The trained ANN predicts waiting time for unseen test data. Regression performance is measured using MAE, RMSE, and R^2. The predicted waiting times are also grouped into Low, Medium, High, and Very High classes to evaluate accuracy, precision, recall, F1-score, and the confusion matrix.

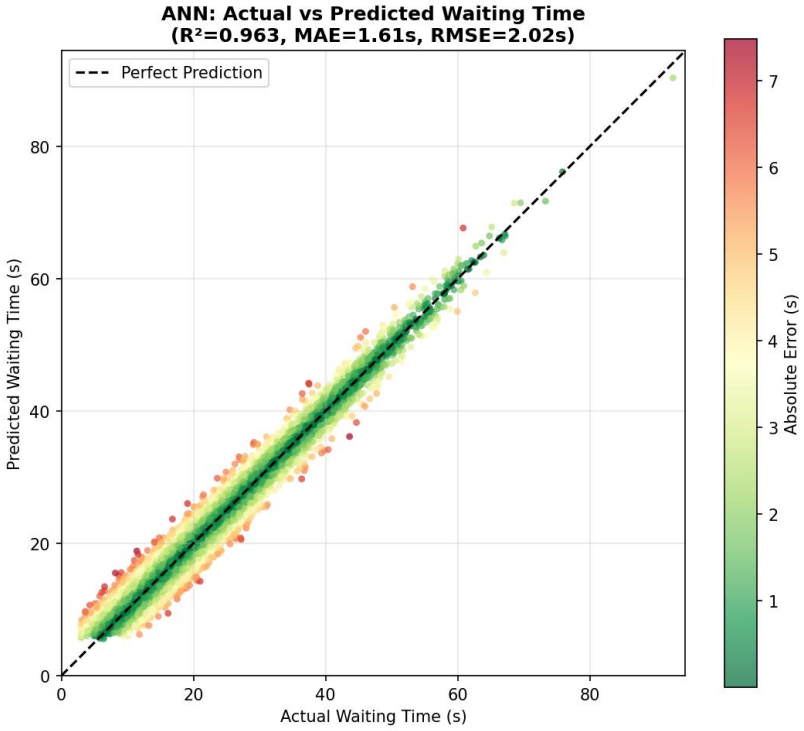


Figure 3: ANN Actual vs Predicted Waiting Time

Figure 3 compares actual and predicted waiting times for the test set. The points are closely distributed around the ideal diagonal line, and the R^2 value of 0.963 indicates strong regression performance. The larger errors are limited mainly to some mid-range waiting-time samples.

Table 1: ANN and System Performance Summary

Metric	Value
Total Records	48,120
Sampled Records (for fuzzy processing)	5,000
Junctions	4
Vehicles Skewness	1.82
ANN MAE	1.61 s
ANN RMSE	2.02 s
ANN R^2	0.9629
ANN Accuracy	0.9763
ANN F1-Score	0.9763
Avg Green Duration (Fuzzy)	56.6 s
Avg Waiting Time (Fuzzy, simulated)	18.2 s
Avg Waiting Time (Fixed-Time, simulated)	23.7 s
Waiting Time Reduction (Fuzzy vs Fixed-Time)	23.0%

Table 1 summarizes the ANN prediction results and the main fuzzy-controller outputs. The ANN gives an MAE of 1.61 s, RMSE of 2.02 s, and R^2 of 0.9629, showing accurate waiting-time estimation. After converting waiting time into classes, both accuracy and F1-score reach 0.9763, meaning that most predictions remain within the correct traffic category.

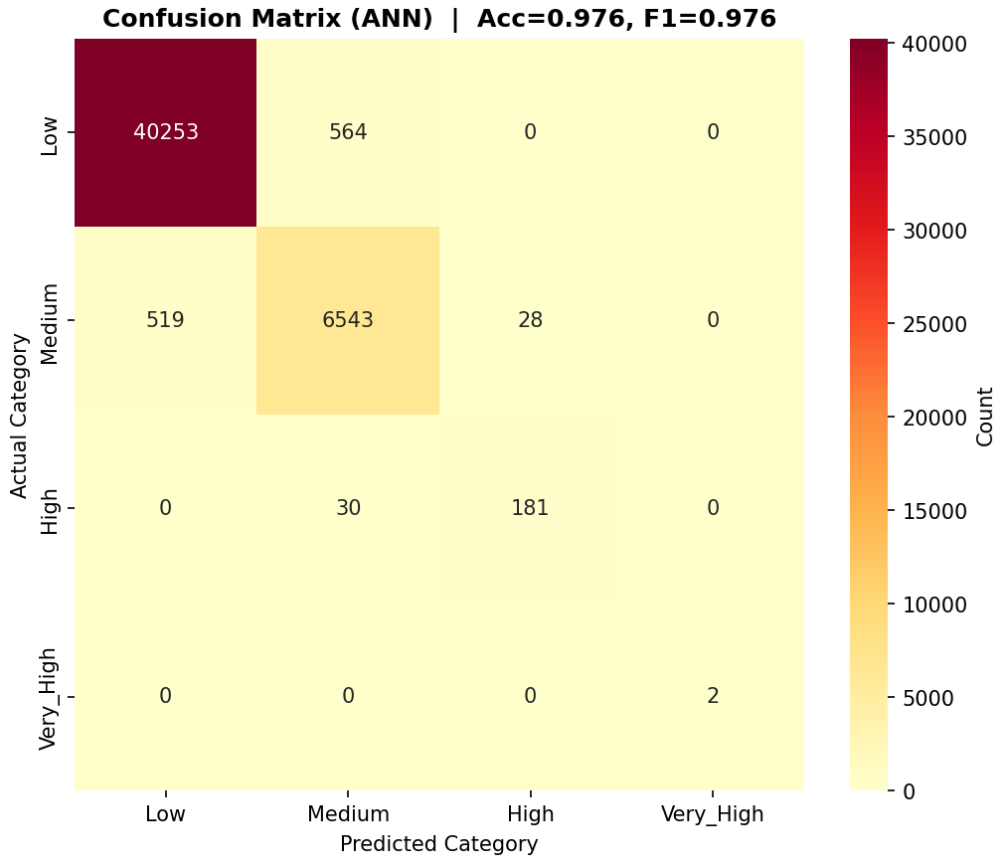


Figure 4: ANN Confusion Matrix

The confusion matrix shows that most records are correctly classified, especially in the Low and Medium categories where most samples lie. The few errors occur mainly between adjacent classes, which is reasonable because waiting time is continuous while the category boundaries are fixed.

2.4 Per-Junction Fuzzy Logic Controller

The ANN-predicted waiting time is supplied to the fuzzy logic controller together with the corresponding traffic density.

Instead of using one controller for the whole network, the proposed system develops a separate fuzzy controller for each junction. The membership functions for traffic density and waiting time are generated from the 5th, 25th, 50th, 75th, and 90th percentiles of each junction, allowing the controller to reflect local traffic conditions.

Each controller uses two input variables:

- Traffic Density
- Predicted Waiting Time

The output variable is:

- Green Signal Duration

For each controller, 25 IF-THEN rules are formed from five traffic-density levels and five waiting-time levels. The fuzzy inference system evaluates these rules and produces an adaptive green duration between 10 s and 90 s according to the traffic condition.

Table 2: Junction-wise Traffic Density and Waiting Time Statistics

Junction	Density Min	Density Q05	Density Median	Density Q90	Density Max	Wait Min	Wait Q05	Wait Median	Wait Q90	Wait Max
1	5.0	15.0	40.0	79.0	156.0	5.4	14.2	28.7	46.9	86.2
2	1.0	6.0	13.0	24.0	48.0	3.0	8.4	16.3	24.0	38.8
3	1.0	3.0	11.0	25.0	180.0	3.0	6.9	16.0	24.5	96.0
4	1.0	2.0	7.0	12.0	36.0	3.0	6.0	13.3	19.9	35.2

Table 2 gives the statistical ranges used for generating fuzzy membership functions. Junctions 1 and 3 have much higher maximum densities than Junctions 2 and 4, showing that a common fuzzy scale would not represent all junctions properly. The per-junction approach therefore provides a better local calibration of density and waiting-time levels.

2.5 Adaptive Signal Timing

After the ANN predicts waiting time, the predicted value is combined with current traffic density and passed to the fuzzy controller assigned to that junction.

For every traffic record, the system performs the following operations:

- The trained ANN predicts the vehicle waiting time.
- The traffic density and predicted waiting time are provided as inputs to the fuzzy controller of the corresponding junction.
- The fuzzy inference system evaluates the predefined rule base.
- The controller calculates the optimal green signal duration for that traffic condition.

Unlike a fixed-time controller, the proposed system changes the green duration according to traffic demand. Higher-density conditions receive longer green time, while low-density conditions avoid unnecessary signal delay.

For computational efficiency, 5,000 representative records are processed through the fuzzy controller. The resulting green durations are then used for analysis and comparison.

2.6 Performance Evaluation

The system is evaluated in terms of ANN prediction performance and traffic-signal control performance.

The ANN is evaluated using regression metrics and an actual-versus-predicted scatter plot. Waiting times are also classified into traffic categories, and the classification output is evaluated using accuracy, precision, recall, F1-score, and the confusion matrix.

The fuzzy controller is evaluated using response curves, sensitivity heatmaps, and a green-duration category breakdown. These outputs show how green time changes with traffic density and waiting time for each junction.

Finally, the adaptive fuzzy controller is compared with a fixed-time controller. The simulated waiting time for both methods is calculated using the same relation, simulated wait = C x density / green duration, where C is calibrated from the dataset. Therefore, the comparison mainly reflects the effect of changing green duration.

SYSTEM ARCHITECTURE

The overall workflow of the proposed intelligent traffic signal control system is shown in the block diagram below.

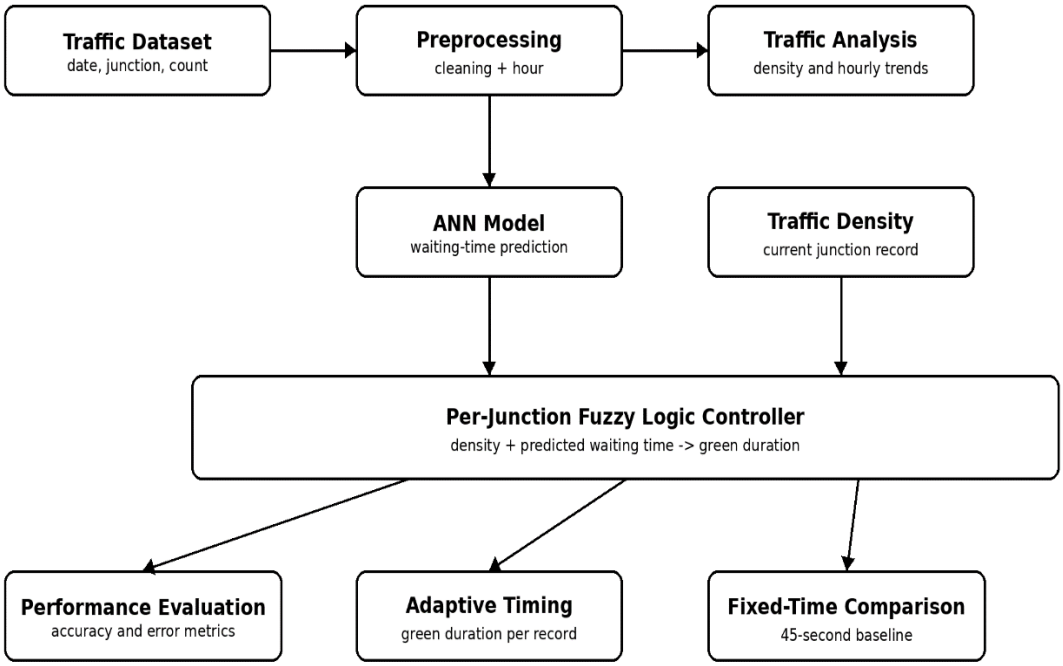


Figure 5: Overall System Architecture

Figure 5 presents the system architecture. The dataset is cleaned and analyzed, after which the ANN predicts waiting time from traffic features. The predicted waiting time and current traffic density are processed by the per-junction fuzzy controller to obtain adaptive green duration. The final stage evaluates the controller and compares it with a 45-second fixed-time signal.

RESULTS AND DISCUSSION

The proposed system combines ANN-based prediction with fuzzy logic control to generate adaptive traffic signal timings. The results show that the model can estimate waiting time and adjust green duration according to traffic conditions.

The exploratory analysis shows that traffic density is unevenly distributed, with low and moderate traffic occurring more often than heavy congestion. The hourly analysis identifies morning and evening peak periods, and the same general pattern appears across junctions with different magnitudes.

The ANN learns the relationship among traffic density, junction number, hour of day, and waiting time. The regression metrics and actual-versus-predicted plot show reliable prediction performance, while the classification metrics show that the predicted waiting times are categorized accurately.

Table 3: Per-Junction Summary of Traffic Density, Waiting Time, and Green Signal Duration

Junction	Records	Avg Density	Min Density	Max Density	Avg Wait (s)	Min Wait (s)	Max Wait (s)	Avg Green (s)	Min Green (s)	Max Green (s)
1	1512	43.69	7.0	122.0	29.81	6.5	69.39	55.65	15.69	85.0
2	1546	14.19	1.0	44.0	16.59	3.0	36.09	57.11	15.90	85.0
3	1474	13.53	1.0	173.0	16.44	3.0	92.62	56.55	15.83	85.0
4	468	7.36	1.0	33.0	13.49	3.0	27.15	57.94	22.52	85.0

Table 3 summarizes traffic density, waiting time, and green duration for each junction. Junction 1 has the highest average density and waiting time, while Junction 4 has the lowest values and the fewest sampled records. Even with these differences, the average green duration remains close across junctions, showing that the per-junction fuzzy controllers normalize local traffic scales effectively.

The fuzzy controller converts predicted waiting time and traffic density into adaptive green signal duration. Since traffic distributions vary between intersections, quantile-based membership functions are generated separately for each junction. This improves local control compared with a single global fuzzy controller.

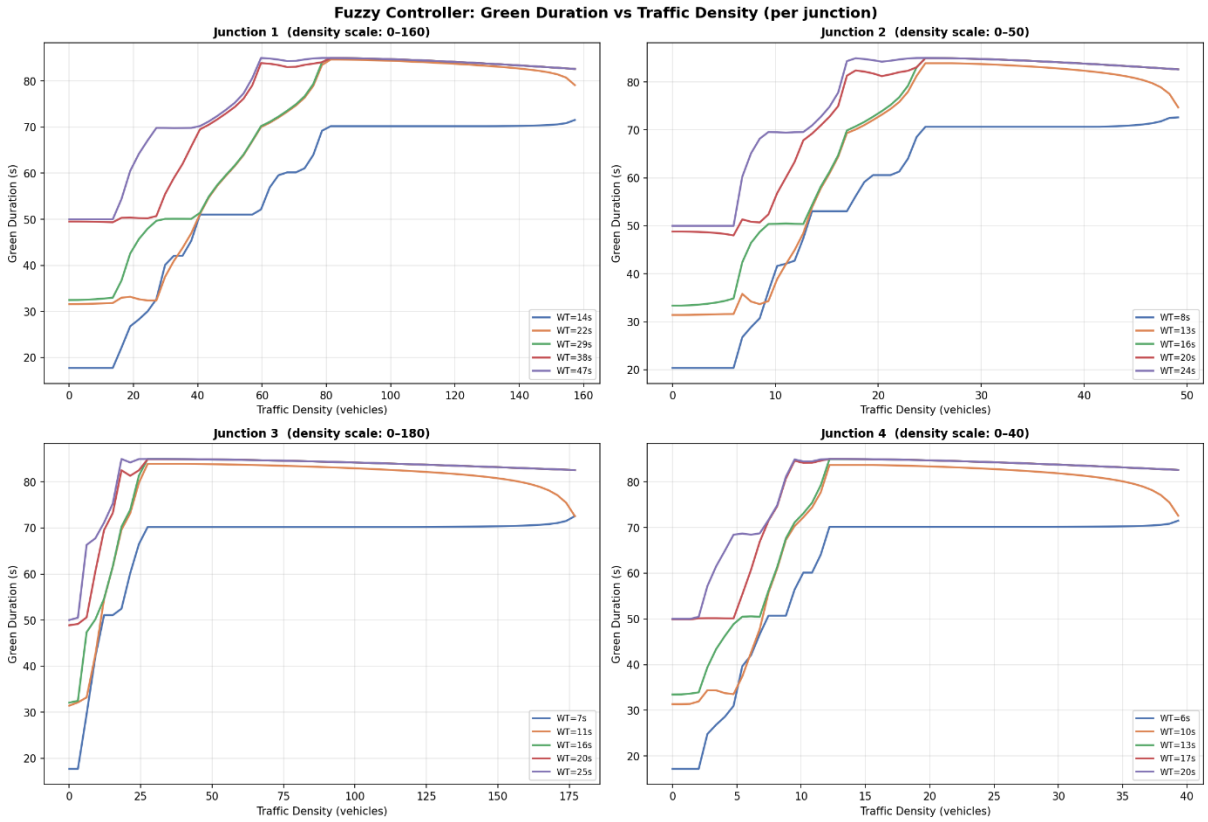


Figure 6: Per-Junction Fuzzy Controller Response — Green Duration vs Traffic Density

Discussion

Figure 6 shows how green duration changes with traffic density for different waiting-time levels. The curves generally increase from a low-duration region to a high-duration region as traffic density rises. Each junction uses its own density scale, so the controller can respond according to local traffic behavior instead of forcing all junctions into one common range.

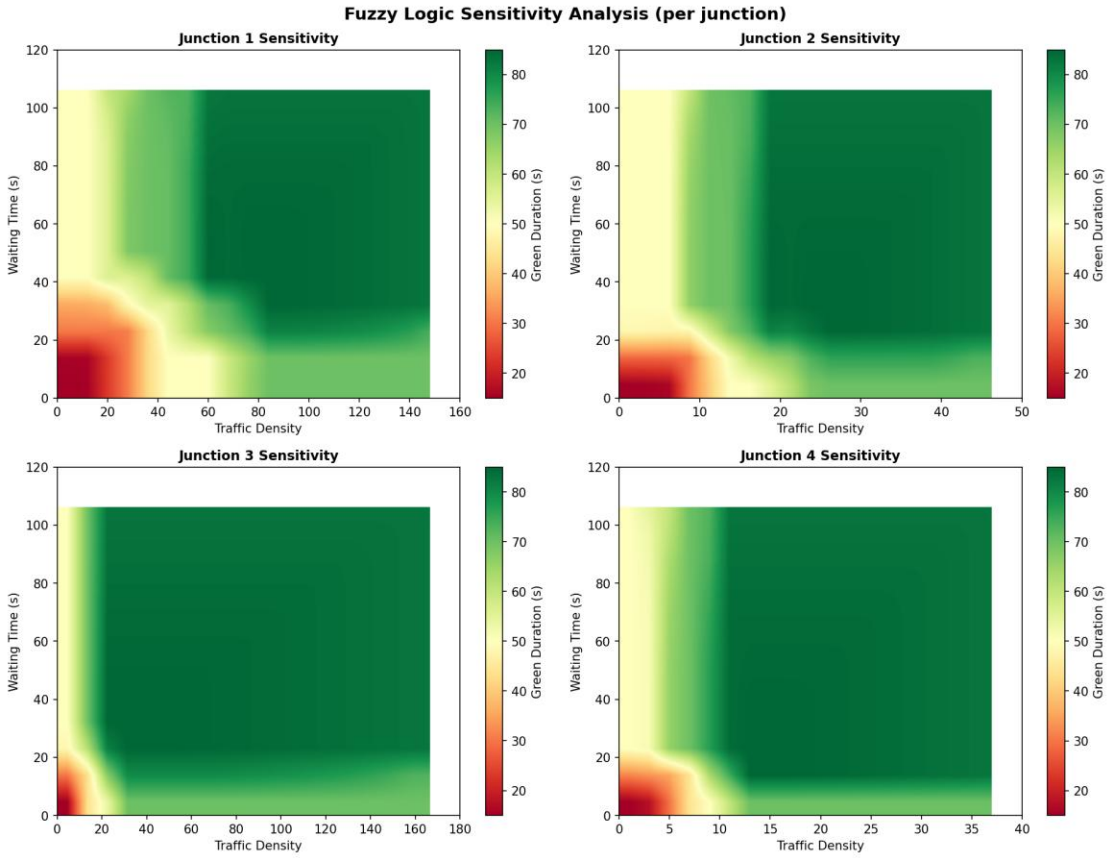


Figure 7: Per-Junction Sensitivity Heatmaps

Discussion

The sensitivity heatmaps show the combined influence of traffic density and waiting time on green duration. Higher density and longer waiting time produce longer green signals, confirming the expected fuzzy-control behavior. The transition region differs by junction because each controller is calibrated using its own traffic scale.

Table 4: Green Light Duration Category Mix by Junction

Junction	Very Short (10-25 s)	Short (25-45 s)	Medium (45-65 s)	Long (65-85 s)
1	4.1%	32.8%	27.9%	35.2%
2	2.1%	29.9%	31.6%	36.5%
3	2.7%	31.3%	30.6%	35.4%
4	1.9%	27.1%	29.3%	41.7%

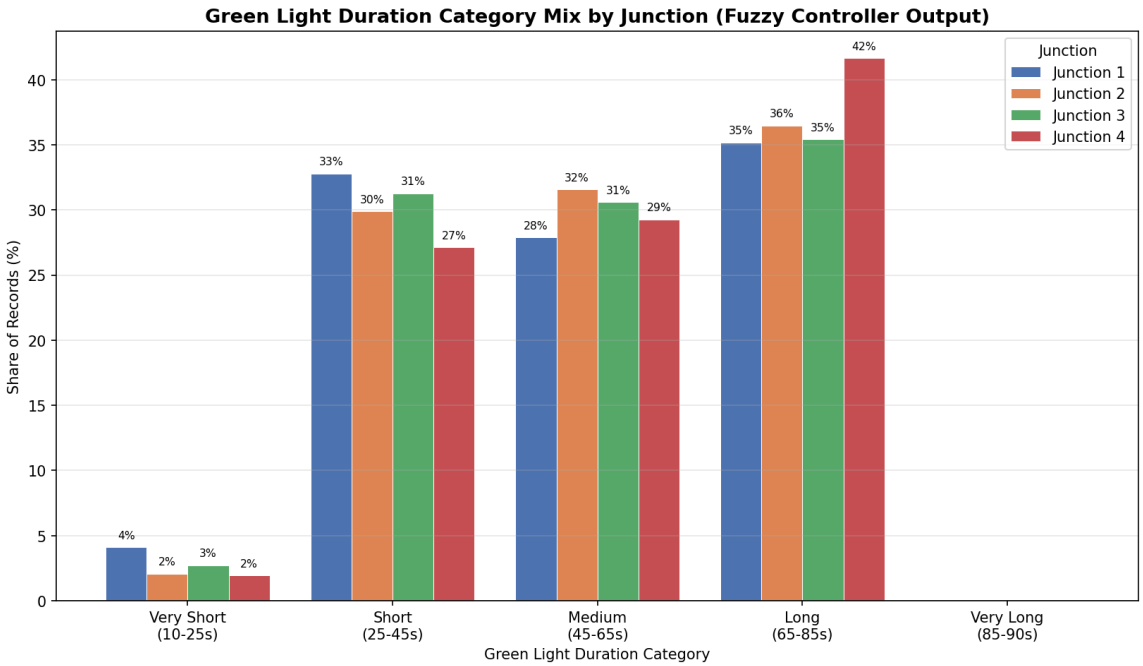


Figure 8: Green Light Duration Category Mix by Junction

Discussion

Figure 8 and Table 4 classify the fuzzy output into green-duration categories. Very Short durations are rare, and most outputs fall in the short, medium, and long ranges. Junction 4 has the highest share of long durations because its local density range is narrower; therefore, a smaller vehicle count can still represent relatively high traffic for that junction.

The fixed-time comparison demonstrates the benefit of adaptive control. Rather than assigning the same green time to every condition, the proposed controller changes the signal duration according to predicted traffic demand.

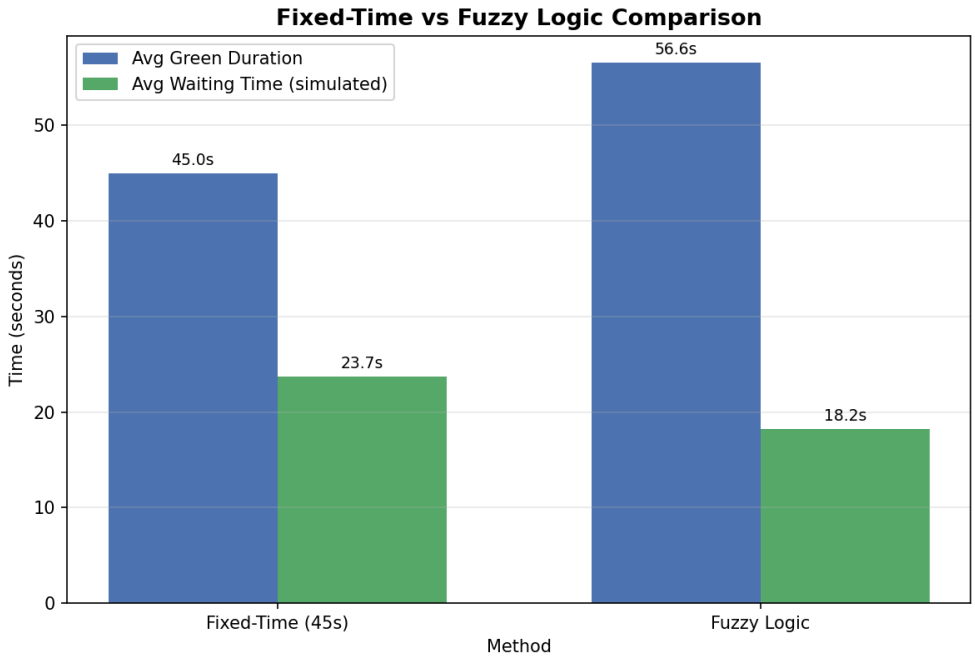


Figure 9: Fixed-Time vs Adaptive Fuzzy Signal Comparison

Discussion

In the calibrated simulation, the fixed 45-second controller gives an average waiting time of 23.7 s. The adaptive fuzzy controller gives 18.2 s, corresponding to a 23.0% reduction, with an average green duration of 56.6 s. Since both methods use the same waiting-time model, the improvement is mainly due to the fuzzy controller allocating green time according to traffic demand.

EVALUATION AND FUTURE ENHANCEMENTS

5.1 Advantages

The proposed system offers the following advantages over conventional fixed-time signal control:

- Dynamically adjusts green signal duration based on real-time traffic conditions.
- Combines ANN prediction with fuzzy logic for intelligent, junction-specific decision-making.
- Reduces waiting time in low traffic and extends green time for congested junctions.
- Enables performance evaluation and comparison with fixed-time signals through visual analysis.
- Extendable to smart city transportation systems.

5.2 Limitations

Despite its effectiveness, the system has the following limitations:

- Waiting time is synthetically generated, not sensor-collected.
- Traffic density is based only on vehicle count; weather, accidents, and pedestrians are not considered.
- No emergency vehicle prioritization.
- Uses historical data rather than live traffic streams; real-time hardware implementation is out of scope.

5.3 Future scope

Future work may extend the system as follows:

- Integrate real-time sensors/CCTV and actual waiting-time data.
- Apply deep learning models (LSTM/GRU) and reinforcement learning for improved, continuously optimized signal prediction.
- Implementation of emergency vehicle and public transport prioritization.

CONCLUSION

This project presents a traffic signal control system that integrates an Artificial Neural Network with a per-junction Fuzzy Logic Controller. The ANN predicts vehicle waiting time using traffic density, junction number, and time of day, while the fuzzy controller calculates green signal duration according to the predicted traffic condition.

The major contribution is the use of independent fuzzy controllers for each junction. Since the membership functions are generated from local junction data, the controller can adapt better than a common fuzzy controller that uses the same scale for all intersections.

The traffic analysis, ANN metrics, confusion matrix, fuzzy response curves, sensitivity heatmaps, category breakdown, and fixed-time comparison show that the proposed system is a practical adaptive traffic-control approach. On the evaluated dataset, the system reduces simulated average waiting time by about 23% compared with the fixed-time baseline. This demonstrates the potential of combining machine learning and fuzzy logic for smart transportation applications.

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